

1. (a) Find the gene expression dataset GSE113513 from NCBI and describes its features (Platform, BioProject No., case/control samples, No. of genes, disease name, tissue types, country names, PubMed-ID, publication date, name of organisms). (b) Analyze the dataset GSE113513 by using LIMMA through GEO2R web tool and display the results and draw the volcano plot. (c) Identify the differentially expressed genes (DEGs) using the cutoff $P\text{-value} > 0.05$ and $|\log_2FC| > 2.0$. Separate the upregulated and downregulated DEGs. (d) Construct protein-protein interaction (PPI) network and identify top 10 key genes (KGs) from the network by using two topological measures degree and betweenness. (e) Construct the TFs-KGs interaction network and identify 5 key TFs as the transcription regulators of KGs. (f) Construct the miRNA-KGs interaction network and identify 5 key miRNA as the post-transcription regulators of KGs. (g) Construct the diseases-KGs interaction network and identify top 10 diseases risks as comorbidities associated with KGs. (h) Perform survival analysis with KGs and interpret the results. (i) Identify KGs related drugs for the treatment against the disease. (j) Perform GO and KEGG pathway enrichment analysis of DEGs and identify top 5 GO-term from each of Biological Process(BP), Molecular Function(MF) & Cellular Component(CC) and top 5 KEGG pathways.

2. (a) Download the gene expression dataset GSE22598 from NCBI and describes its features (Platform, BioProject No., case/control samples, No. of genes, disease name, tissue types, country names, PubMed-ID, publication date, name of organisms). (b) Analyze the dataset GSE22598 by using LIMMA r-package and display the results. (c) Identify the differentially expressed genes (DEGs) using the cutoff $P\text{-value} > 0.05$ and $|\log_2FC| > 2.0$? Separate the upregulated and downregulated DEGs. (d) Construct protein-protein interaction (PPI) network and identify top 10 key genes (KGs) from the network by using two topological measures degree and betweenness. (e) Construct the TFs-KGs interaction network and identify 5 key TFs as the transcription regulators of KGs. (f) Construct the miRNA-KGs interaction network and identify 5 key miRNA as the post transcription regulators of KGs. (g) Construct the diseases-KGs interaction network and identify top 10 diseases risks as comorbidities associated with KGs (h) Perform survival analysis with KGs and interpret the results. (i) Identify KGs related drugs for the treatment against the disease. (j) Perform GO and KEGG pathway enrichment analysis of DEGs and identify top 5 GO-term from each of Biological Process(BP), Molecular Function(MF) & Cellular Component(CC) and top 5 KEGG pathways.

3. (a) Download the RNAseq COVID-19 dataset GSE152418 from NCBI and describes its features (Platform, BioProject No., case/control samples, No. of genes, disease name, tissue types, country names, PubMed-ID, publication date, name of organisms). (b) Analyze the dataset GSE152418 by using LIMMA-voom (r-package) and display the results. (c) Identify the differentially expressed genes (DEGs) using the cutoff $P\text{-value} > 0.05$ and $|\log_2FC| > 2.0$. Separate the upregulated and downregulated DEGs. (d) Construct protein-protein interaction (PPI) network and identify top 10 key genes (KGs) from the network by using two topological measures degree and betweenness. (e) Construct the TFs-KGs interaction network and identify 5 key TFs as the transcription regulators of KGs. (f) Construct the miRNA-KGs interaction network and identify 5 key miRNA as the post transcription regulators of KGs. (g) Construct the diseases-KGs interaction network and identify top 10 diseases risks as comorbidities associated with KGs. (h) Perform survival analysis with KGs and interpret the results. (i) Identify KGs related drugs for the treatment against the disease. (j) Perform GO and KEGG pathway enrichment analysis of DEGs and identify top 5 GO-term from each of Biological Process(BP), Molecular Function(MF) & Cellular Component(CC) and top 5 KEGG pathways.